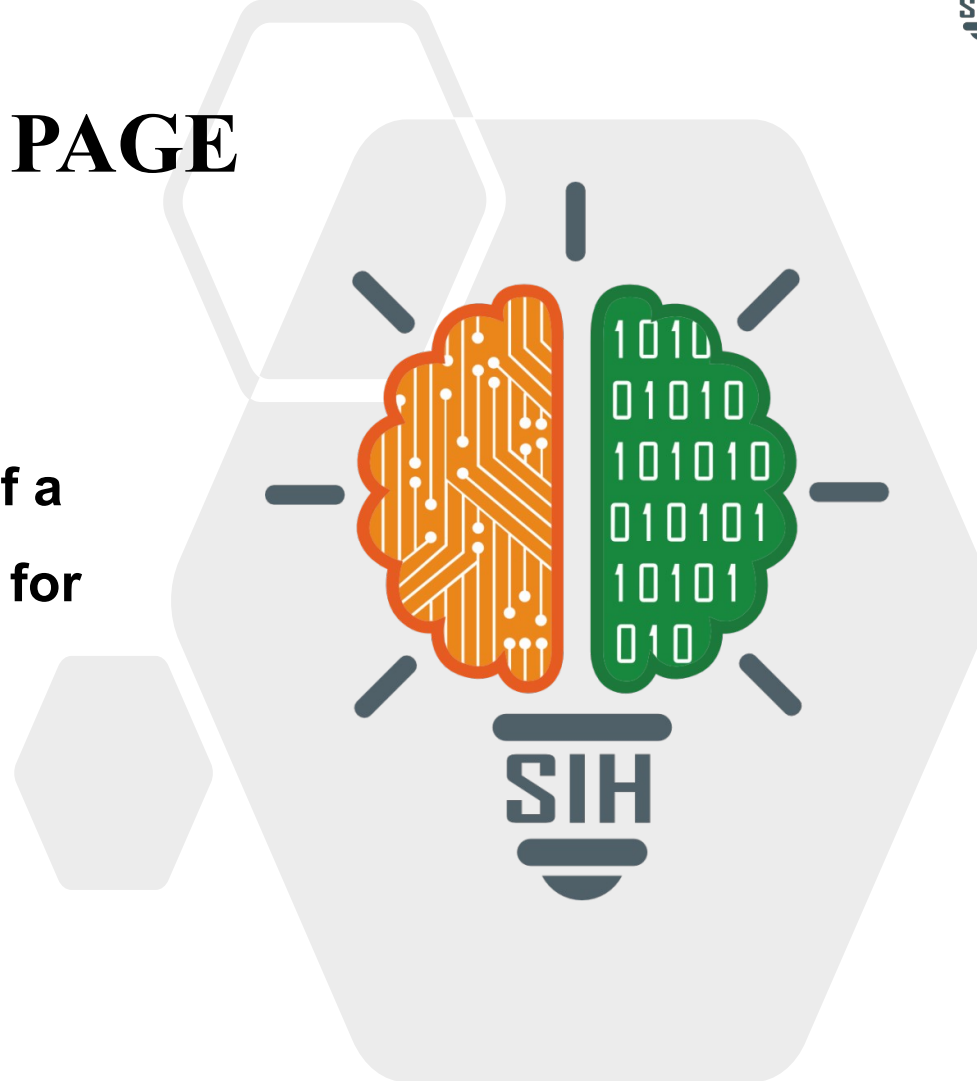
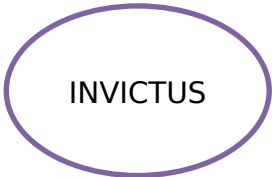




TITLE PAGE

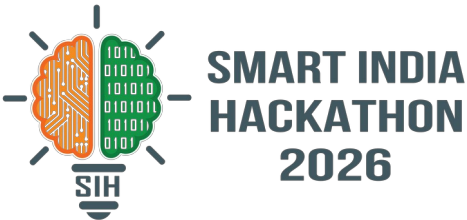
- Problem Statement ID – 26110
- Problem Statement Title- Development of a Low-Cost Light-weight Milk Chilling Can for Small-Scale Dairy Farmers
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Hardware
- Team ID- 159400
- Team Name- INVICTUS



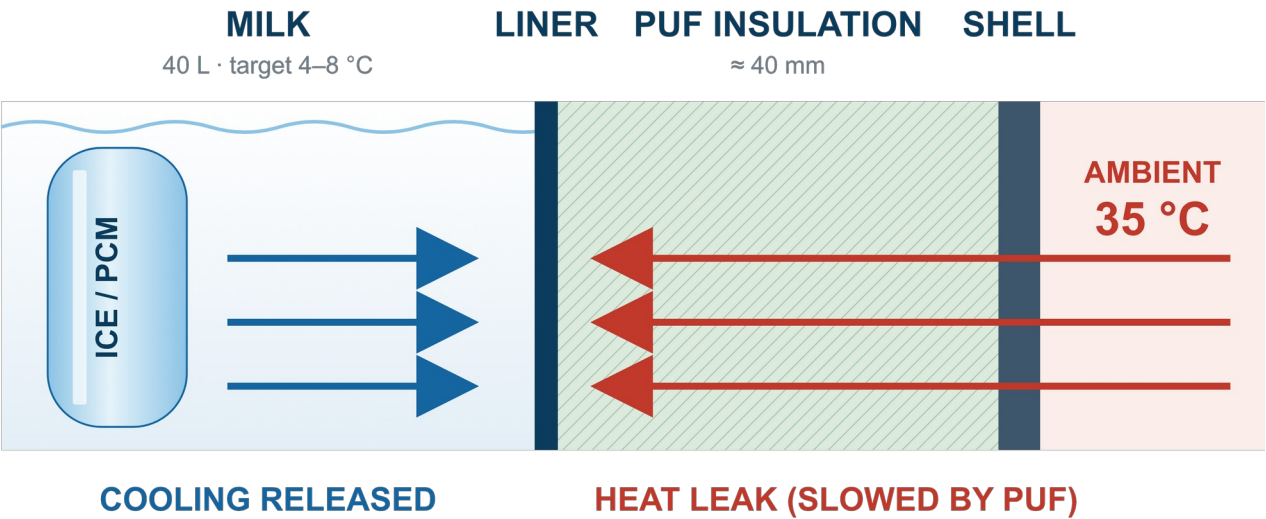


MILKSHIELD

Passive Cooling Milk Chilling Can, 30–40 L



❖ Proposed Solution



30–40 L capacity	4–8 °C milk temperature	6–12 h hold time	0 W continuous power
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PS requirements – simulated: below 8 °C in 75 min, held 14 h at 35 °C

Optional monitoring:



WHAT IT IS

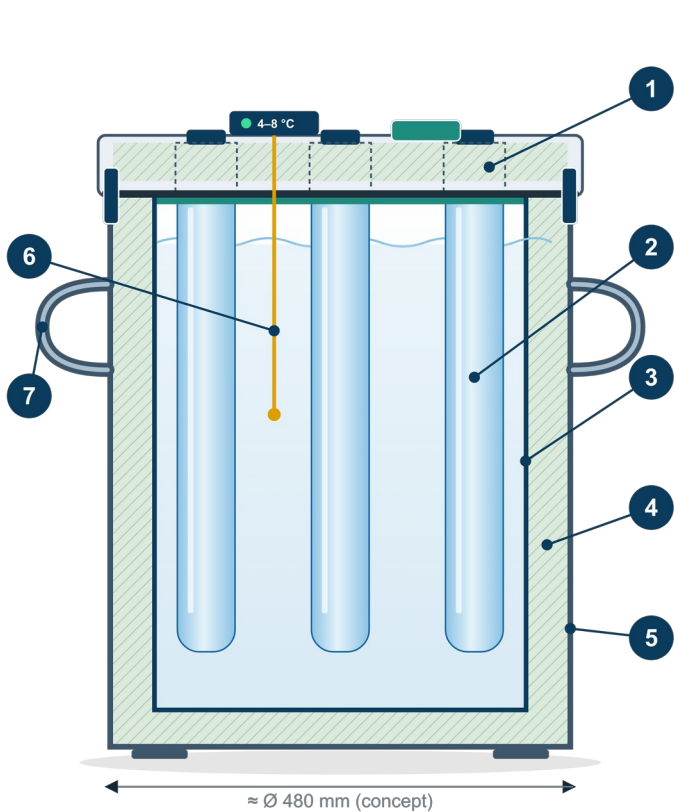
- Double-wall PUF-insulated can with a food-grade HDPE chamber
- Reusable ice/PCM cartridges sit in the milk – no compressor, no mains power

HOW IT WORKS

Passive cooling stores cooling energy in reusable PCM/ice cartridges and releases it gradually around the insulated milk chamber without continuous external power.

WHAT IS NEW

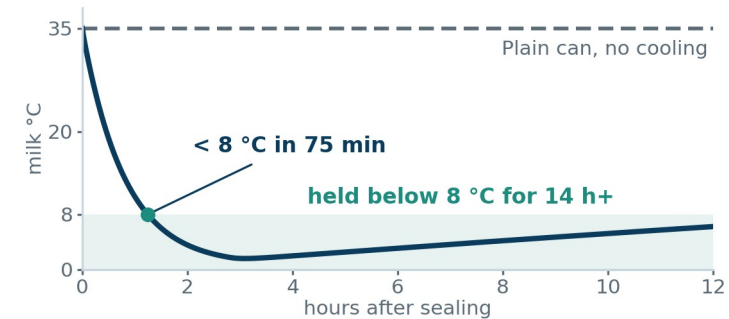
- **Modular cartridges:** capacity set per load
- **Ice inside the milk:** direct, fast heat transfer
- **Easy to clean:** smooth, wide-mouth liner



- Lid + gasket**
Insulated PE lid, food-grade seal, cam latches
- Ice/PCM cartridge**
Sealed 2 L food-grade HDPE, water/ice
- Inner liner**
Food-grade HDPE (IS 10146) [7], smooth
- PUF insulation**
≈40 mm, $k \approx 0.022\text{--}0.035 \text{ W/m}\cdot\text{K}$ [6]
- Outer shell**
UV-stabilised rotomoulded PE
- Sensor (optional)**
Probe → LED; ESP32 logger optional
- Handles**
Moulded-in, low centre of gravity

Made by: rotomould shell + liner → PUF pour → assemble

THERMAL SIMULATION 40 L · 35 °C day

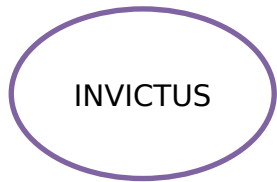


CARTRIDGES NEEDED (2 L each)

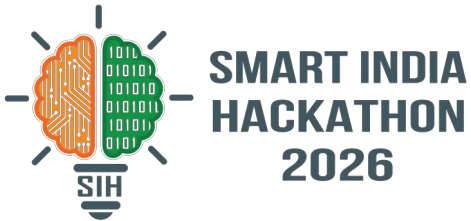
Pre-chilled milk · 12 h	2
Fresh 30 L · 6 h	6
Fresh 40 L · 12 h	8

Fresh-milk pull-down is $\approx \frac{3}{4}$ of the load, so cartridge count sets capacity.

Sized by simulation: 35 °C ambient, 40 mm PUF; milk cp, ρ [8].



FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea	Potential challenges and risks	Strategies for overcoming these challenges
Cooling duration Simulated: 14 h below 8 °C	Fresh milk needs $\approx 3\times$ the cooling of a 12 h hold	Cartridge count sized per load by simulation (2–8)
Heat leakage ≈ 22 W at 35 °C (wall heat-flow model)	Lid joint is the main weak spot (≈ 11 % of leak)	Insulated lid, gasket, no metal through-paths
Weight Manageable	≤ 16 kg of ice for a full 40 L, 12 h trip	Carry only what the load needs; swap at collection centre
Hygiene Food-grade HDPE, easy to clean	Milk residue, little cleaning water	Smooth liner, wide mouth, crown lifts out
Leakage · rough roads Robust	Bumps, tipping, cartridge or lid leaks	Rotomoulded PE, gasket + latches, welded cartridges
Cost · manufacturing Viable	Tooling cost at volume	Prototype from off-the-shelf drums; rotomould + PUF later

TECHNICALLY FEASIBLE
Simulation + prior art [1]

ECONOMICALLY FEASIBLE
Prototype $\approx$ ₹6–13 k (est.)

OPERATIONALLY FEASIBLE
Cartridges swapped at centre

MANUFACTURABLE
Rotomould + PUF process

SCALABLE
Cartridge count sets capacity

TARGET USERS

- 1 Small & marginal dairy farmers
- 2 Village milk collection centres
- 3 Dairy cooperatives
- 4 Rural milk transporters
- 5 Remote / hilly / NE dairy networks



Today: uninsulated cans. Karnal, Haryana. Photo: Arne Hückelheim, [CC BY-SA 3.0](https://commons.wikimedia.org/wiki/File:Arne_Hueckelheim_-_Karnal_Haryana_India_2014_01_01_01.jpg), Wikimedia Commons

Codex: cool to ≤ 6 °C (≤ 4 °C if not collected daily) when not collected within 2 h [4]

BENEFITS

SOCIAL

- Safe milk without electricity
- Suits hilly & NE villages

ECONOMIC

- Reusable ice/PCM, no power bill
- Fewer rejected milk lots

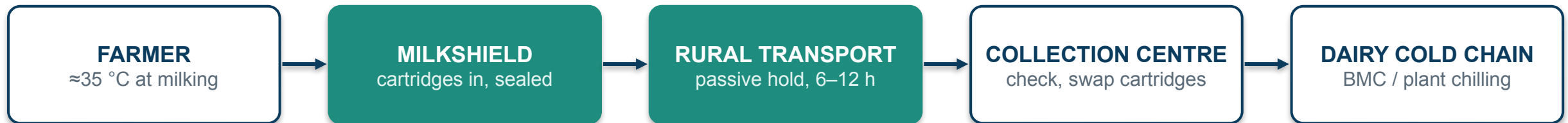
OPERATIONAL

- Easy to clean, moulded handles
- Cartridges sized per trip

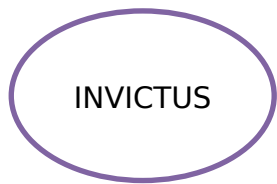
ENVIRONMENTAL

- No refrigerant, no diesel
- Reusable cartridges, recyclable HDPE

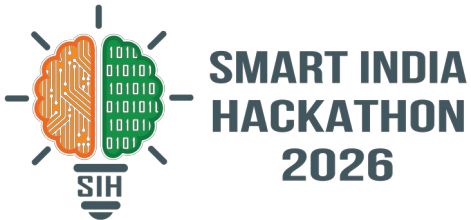
MILK JOURNEY



Scales later with the same design: 30–40 L → 50–100 L modular → cooperative fleets



RESEARCH AND REFERENCES



	Conventional can [5]	Ice-in-can prototype [1]	PCM pail, ICAR-NDRI [2,3,9]	MilkShield (proposed)
Cooling	None	Ice compartment	Nano-PCM jacket	Ice/PCM cartridges in milk
Power	None	None in use	None in use	None in use
Reported result	No cooling	30 L → 17 °C in <90 min; 20 L <13 °C for 12 h	37 → <10 °C in 30–40 min (reported)	Simulated: below 8 °C in 75 min; held 14 h at 35 °C
Modular	No	No	No	Yes: 2–8 cartridges
Cost	≈ ₹2,350 (40 L Al)	Not reported	≈ ₹1,500–2,000	Prototype ≈ ₹6–13 k (est.)

REFERENCES

[1] Torres-Toledo et al., Int. J. Refrigeration 90 (2018) 22–31. [doi:10.1016/j.ijrefrig.2018.04.001](https://doi.org/10.1016/j.ijrefrig.2018.04.001)

[2] Prakash et al., Innov. Food Sci. Emerg. Technol. 81 (2022) 103112. [doi:10.1016/j.ifset.2022.103112](https://doi.org/10.1016/j.ifset.2022.103112)

[3] Prakash & Ravindra, J. Food Eng. 337 (2023) 111223. [doi:10.1016/j.jfoodeng.2022.111223](https://doi.org/10.1016/j.jfoodeng.2022.111223)

[4] FAO/WHO Codex, Code of Hygienic Practice for Milk and Milk Products, CAC/RCP 57-2004

[5] IGNOU / eGyanKosh, Dairy Technology Unit 1: Milk Collection & Transportation

[6] Verma & Singh, Energy Procedia 161 (2019) 232–241. [doi:10.1016/j.egypro.2019.02.086](https://doi.org/10.1016/j.egypro.2019.02.086)

[7] FSSAI Packaging Regulations 2018 (IS 10146); BIS IS 1825:1983

[8] Minim et al., J. Chem. Eng. Data 47 (2002) 1488. [doi:10.1021/je025546a](https://doi.org/10.1021/je025546a)

[9] DST, Govt. of India: affordable milk chilling at point of harvest

[10] Project repository: github.com/Sanmugavel1/MilkShield-SIH2026

Simulation study: thermal model · wall heat-flow model · 2D CFD (milk cools evenly, top to bottom within 2.5 °C). Next: prototype field trial.
Full simulation code, results & videos: github.com/Sanmugavel1/MilkShield-SIH2026